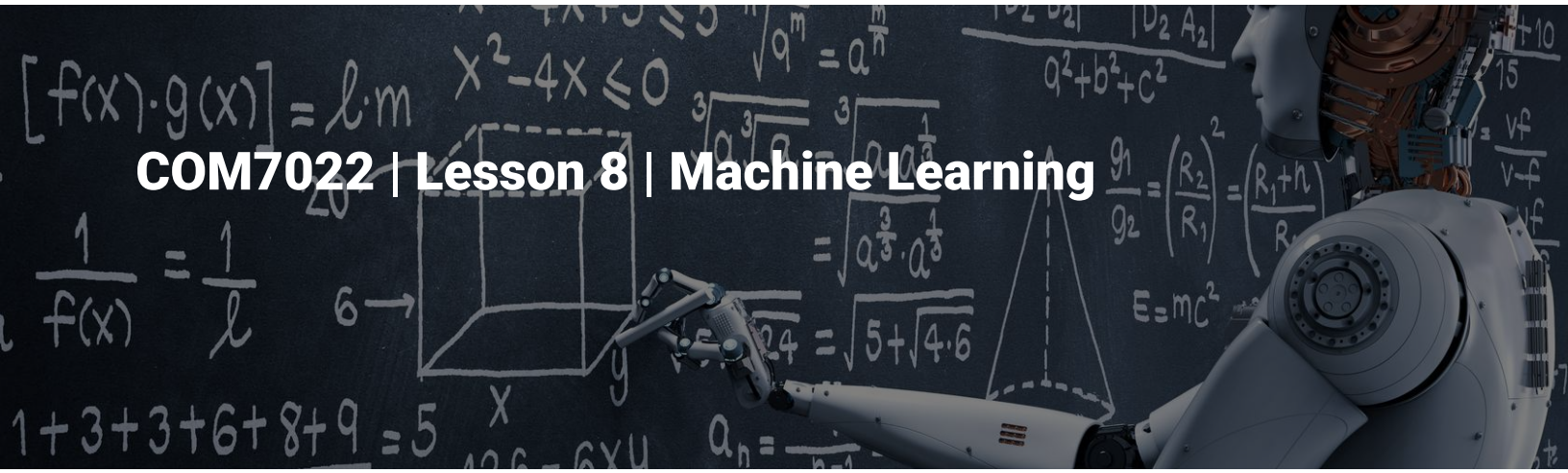

COM7022 | Lesson 8 | Machine Learning

Machine Learning (COM7022)

Lesson 8: Clustering

Cover image by Phonlamai Photo from Shutterstock.com

- **Lesson introduction**
- **Before you get started**
- **Topic 1: What is Cluster Analysis?**
- **Topic 2: Type of Cluster techniques**
- **Topic 3: K-Means Clustering**
- **Topic 4: Building and evaluating a K-Means Clustering model**
- **Knowledge check**

Lesson summary

Reading List

Lesson introduction

Welcome to Lesson 8

LESSON INTRODUCTION | LEARNING OUTCOMES | ACCESSIBILITY GUIDANCE

Welcome to Lesson 8!

Lesson 8 has been designed to take you through the complexities of cluster analysis. We will begin the lesson by defining what cluster analysis is and the differing techniques that are available to the analyst, dependent upon data environments. Once the theory has been detailed, the lesson focuses in on one particular and arguably the most common cluster model, K – Means Clustering. In the final topic, we will be giving an in-depth example on how to apply a K – Means Cluster algorithm to a targeted dataset.

As you move through your professional career, you will no doubt engage in one, if not all, of the common topics detailed within this lesson.

Computing

At Arden University

LESSON INTRODUCTION | LEARNING OUTCOMES | ACCESSIBILITY GUIDANCE

By the end of this week's lesson you will:

- Know what cluster analysis is used for and, more importantly, why.
- Understand the differences between cluster analysis techniques.
- Have a clear definition of what K-Means Clustering is and appreciate why it is commonly used.
- Be experienced in applying a K – Means analysis to a targeted dataset, plus assess accuracy.

Learning Outcomes

Lesson 8 aligns to Learning Objectives 1 and 3:

- **Demonstrate a practical understanding of the characteristics of various Machine Learning methods and their applications in data analysis.**
- **Critically evaluate the results from the application of Machine Learning techniques and propose improvements.**

Computing
At Arden University

We have worked across teams at Arden University to try and create an accessible and inclusive experience within the learning content. We are continuously assessing the accessibility of our lessons to make improvements and enhancements based on testing and feedback.

This lesson may include screenshots of code and software used. Please be aware that these may not always have a text alternative, and that other features such as size and colour may not be accessible.

If you experience any issues with accessing the learning content throughout this module, there are several support services in place to help you.

- Arden University has a dedicated Inclusion Services Team who can help you with getting adjustments and equipment.
- The Library Portal also has information on the accessibility tools that can help you access resources: Accessibility Tools – Library Portal
- If you are struggling with accessing a specific part of the content e.g. a video resource that is missing captions, please speak to your Lecturer who may be able to advise on an alternative resource in the interim to ensure you are able to get the learning you need to progress with the lesson.

Continue to the Before you get started section

Before you get started

Image from Articulate Rise 360 Content Library.

To give cluster analysis context and background knowledge within Machine Learning environment, please read through and conduct the examples in Chapter 8: Data Science (pp. 127-144) from Python for Scientific Computing and Artificial Intelligence by Lynch.

Continue to Topic 1

Topic 1: What is Cluster Analysis?

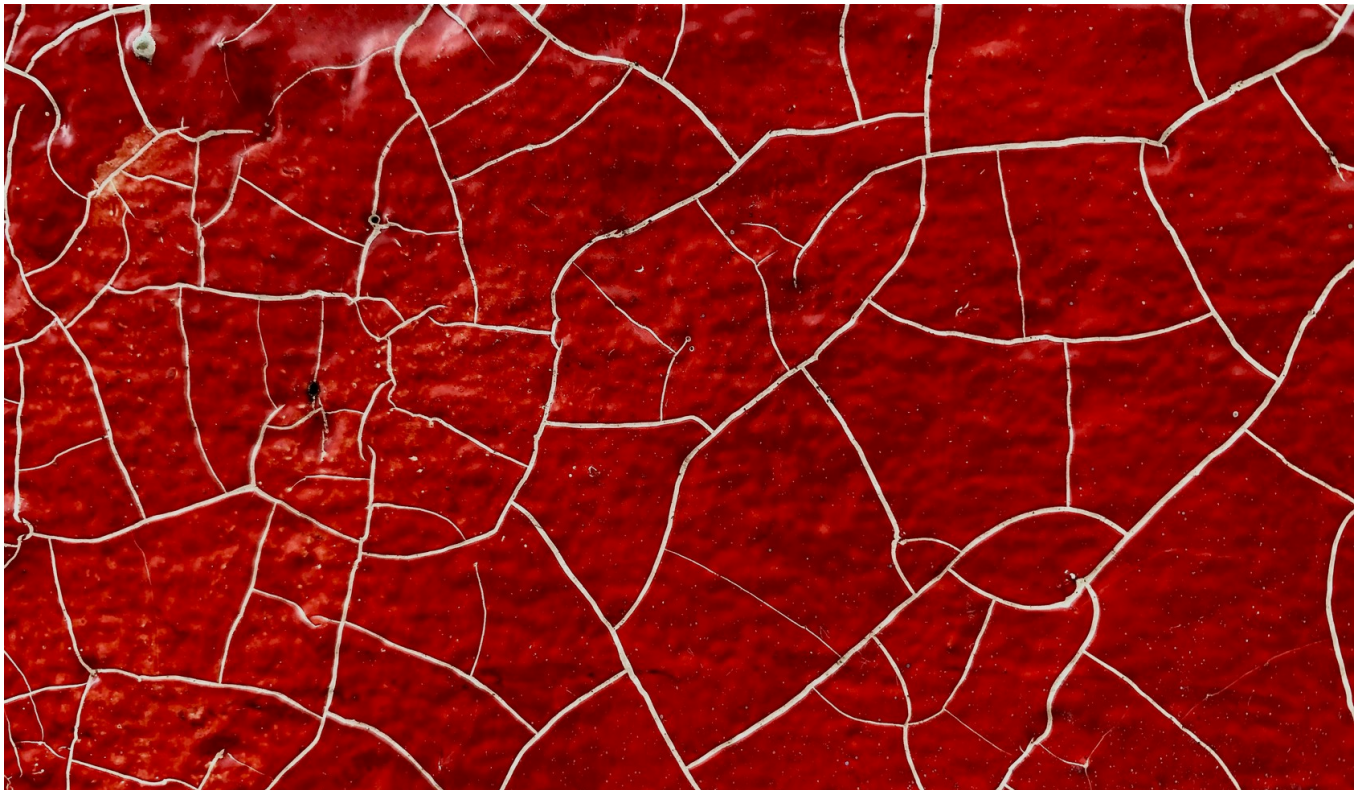

Image from Articulate Rise 360 Content Library.

Let us begin this module with the following statement,

> *"Whatever machine learning investigation you will be doing, both now and in the future, somewhere along the lines you will be using some form of Cluster analysis."*

A bold statement I agree, but nevertheless, especially when analysing large, unknown datasets, it will undoubtable be the case. The term "unknow" here refers to unsupervised learning, which cluster analysis falls within.

Cluster analysis is a straightforward way for an analysist to disseminate large, unknow and unlabelled datasets. The technique is especially useful to market analysists, concerning customer geographical locations and for insurance analysists, where risk can be conducted relatively easily on new and existing customers.

Cluster analysis, however, on its own is not the solution in many machine learning environments, it would however be part of an initial investigation. Once initially deployed, it gives the analysist guidance on how to further study the dataset, so that a more in-depth investigation can be undertaken.

Quite frequently within a machine learning environment you will hear the term "Cluster analysis", but what is it exactly? An important question and indeed warrants an in-depth explanation. In essence, the technique identifies groups of data that share some similarity. What those similarities are is initially not known, which is why further study is required to extract meaning results. Nevertheless, cluster analysis is an ideal machine learning tool to initially use when an analysis has no, or little idea of what differing data types are included within a large unlabelled dataset.

As I alluded to earlier, cluster analysis does have its limitations and these are primarily based upon differing strategies used by the analysts, as these are sometimes not related to the data itself, but its positioning within neighbouring data points. Take for example K – Means clustering, where datasets are split, or grouped dependent upon how many centroids the machine analyst has

initially “guessed”. As can be no doubt be imagined, the initial centroid “guess” can have quite an impact on the results of the investigation. Of course, there are further techniques which can be used to limit the centroid guesswork and this will be covered later in the module, but the initial issue is still there. Hierarchical clustering on the other hand is initially more accurate, as we shall see further on in this module, however, the caveat to this is that it requires far more computer power and therefore not great when working on large datasets. Plus, Hierarchical clustering is highly sensitive to outliers within a dataset, thus has the tendency to produce inaccurate initial results, therefore care must be taken.

As I alluded to earlier, cluster analysis does have its limitations and these are primarily based upon differing strategies used by the analysts, as these are sometimes not related to the data itself, but its positioning within neighbouring data points. Take for example K – Means clustering, where datasets are split, or grouped dependent upon how many centroids the machine analyst has initially “guessed”. As can be no doubt be imagined, the initial centroid “guess” can have quite an impact on the results of the investigation. Of course, there are further techniques which can be used to limit the centroid guesswork and this will be covered later in the module, but the initial issue is still there. Hierarchical clustering on the other hand is initially more accurate, as we shall see further on in this module, however, the caveat to this is that it requires far more computer power and therefore not great when working on large datasets. Plus, Hierarchical clustering is highly sensitive to outliers within a dataset, thus has the tendency to produce inaccurate initial results, therefore care must be taken.

Hopefully, you can determine that cluster analysis should not be used as a single point of investigation. However, it is a very good and highly adopted initial investigation tool that can be used to initially separate unknown, unlabelled data for a more in-depth and focused machine learning progressive study.

Continue

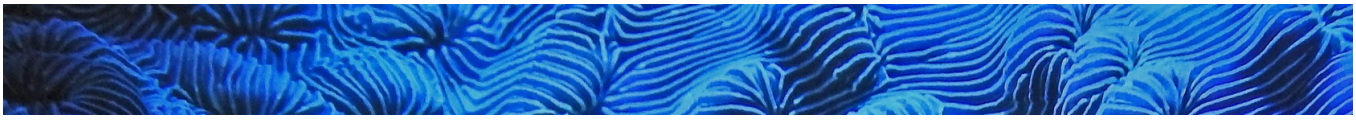

Image from Articulate Rise 360 Content Library.

First and foremost, a clustering should be adopted by the machine learning analyst where it will have the biggest impact and initial insight: usually upon large, unknown datasets that have been within a company's data warehouse for some time. Therefore, if used in this manner, it can reduce the workload of the investigation considerably. Similar to other classification algorithms, clustering can take large datasets, without much initial input from the analysist and efficiently group data points together to make the dataset more useable. The biggest advantage to cluster analysis is that it can give quick, initial answer on huge datasets, thus driving further direction.

Another big advantage of using cluster analysis is its ability to quickly identify outliers or anomalies, which was mentioned earlier on. Take the Density Based Spatial Clustering of Applications with Noise (DBSCAN) clustering algorithm: the algorithm is specifically designed to search for clusters that are closely positioned to each other, which in turn will identify outlier's easily: hitherto making more accurate results.

When used in the correct context within a data study, cluster analysis is a powerful technique which a machine learning analysist can have in their arsenal. The technique can quite quickly and

efficiently group large, unlabelled datasets, based upon similarities within individual data points. In doing so, it aids the analyst in direction for further, more focused investigations, therefore extracting meaningful data from what was originally messy, dark data.

Continue

Topic 1 - Activity 1

Discussion on iLearn

Please conduct further research on Cluster analysis and state in your own words why you would apply the algorithm to a data investigation and at what point. Also detail why a cluster analysis might be important when dealing with unlabelled data. Place your answer in your group's discussion forum for further discussions with peers and tutors.

Continue

Topic 1 - Activity 2

Reading

- Chapter 3: Unsupervised Learning and Preprocessing (pp. 103-151) from Introduction to Machine Learning with Python: A Guide for Data Scientists by Müller and Guido.
- Chapter 3: The Data Effect: A Glut at the End of the Rainbow (pp. 126-145) from Predictive Analytics: The Power to Predict Who Will Click, Buy, Lie, or Die, Revised and Updated by Siegel.

Continue

TOPIC 1 SUMMARY

Well done, you have completed Topic 1.

Within this topic we have detailed what clustering is and why it is used so extensively within a machine learning environment. Hopefully, you now fully understand that although cluster analysis would not be used as a standalone technique, it is a very good starting point on which to impose upon unlabelled datasets. In using a cluster analysis as an initial starting point, it gives analysts some guidance on the next steps to take for a more focused study.

The initial theory and reasoning completed; we can move onto detailing various other forms of clustering algorithms available to a machine learning study.

See you in the next topic.

Continue to Topic 2

Topic 2: Type of Cluster techniques

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As we alluded to within the first topic, there are many different types of cluster analysis available and we shall separately discuss the most common forms below.

Continue

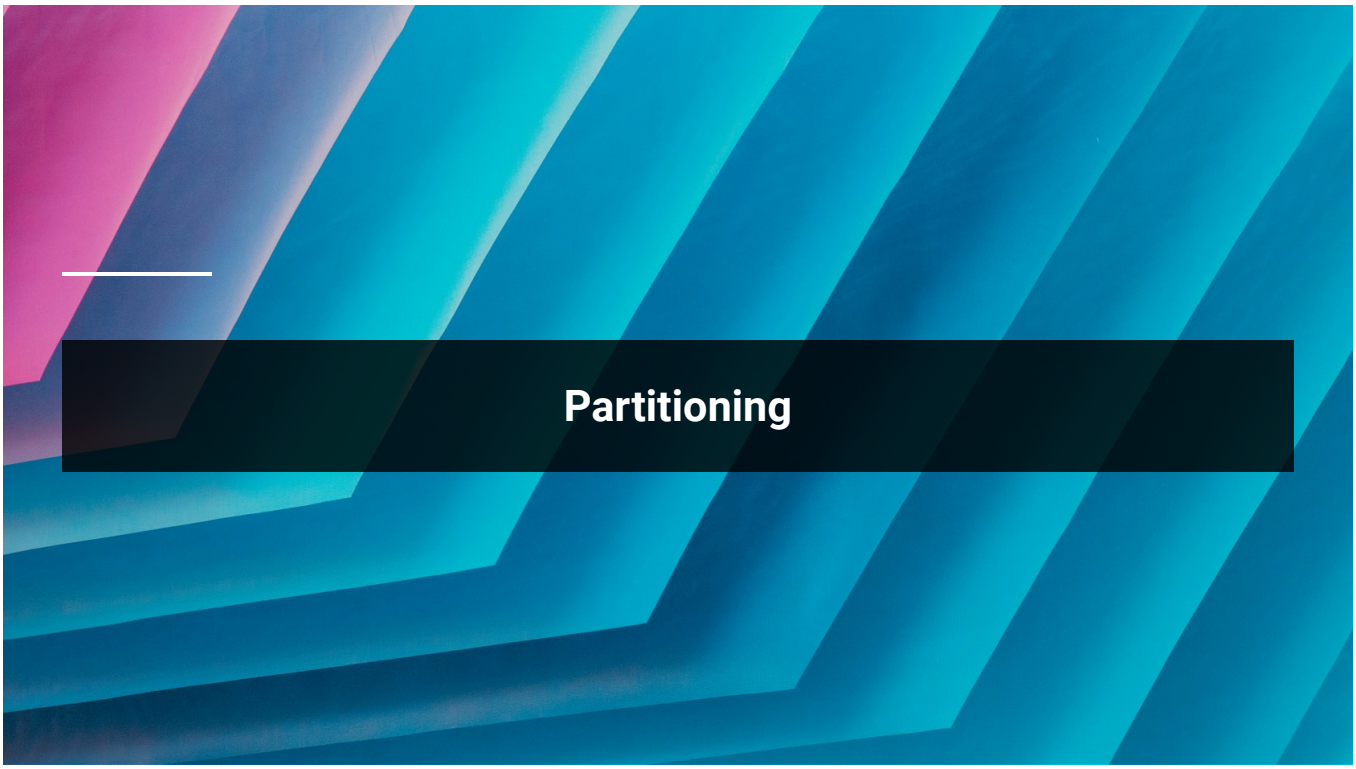


Image from Articulate Rise 360 Content Library.

Partitioning is widely regarded as the backbone of modern-day search engines. When a search is undertaken, the search results are grouped together to give associated results. This is exactly what is going on if you perform a new genre search in Netflix, based upon past history.

Primarily, partitional clustering groups, or partitions data into K distinct clusters, grouping similar data points together, thus maximising the differences between each cluster. The K represents the number of clusters used: as this could be any number, we define that algebraically as K. The most used partition clustering technique used in data science in general is K – Means clustering: a technique we shall be looking and giving an in-depth example on in topic s 3 and 4 of this lesson. There are however other partitional clustering techniques available, such as Fuzzy C- Clustering and Hierarchical clustering. The latter of which we shall be taking a deep dive into in Lesson 9 of this module.

Fuzzy C – Means Clustering is a form of “soft” clustering, which means that each data point is separated into different clusters and then given a probability being assigned that cluster. Hard clustering however, works on the principle that a particular data point belongs to a cluster, or not. Therefore, the advantage to a soft clustering technique is that it has the potential to give better results where overlapping data points are concerned, when compared to hard clustering. Figure 8.01 below graphically demonstrates the difference between hard and soft clustering techniques.

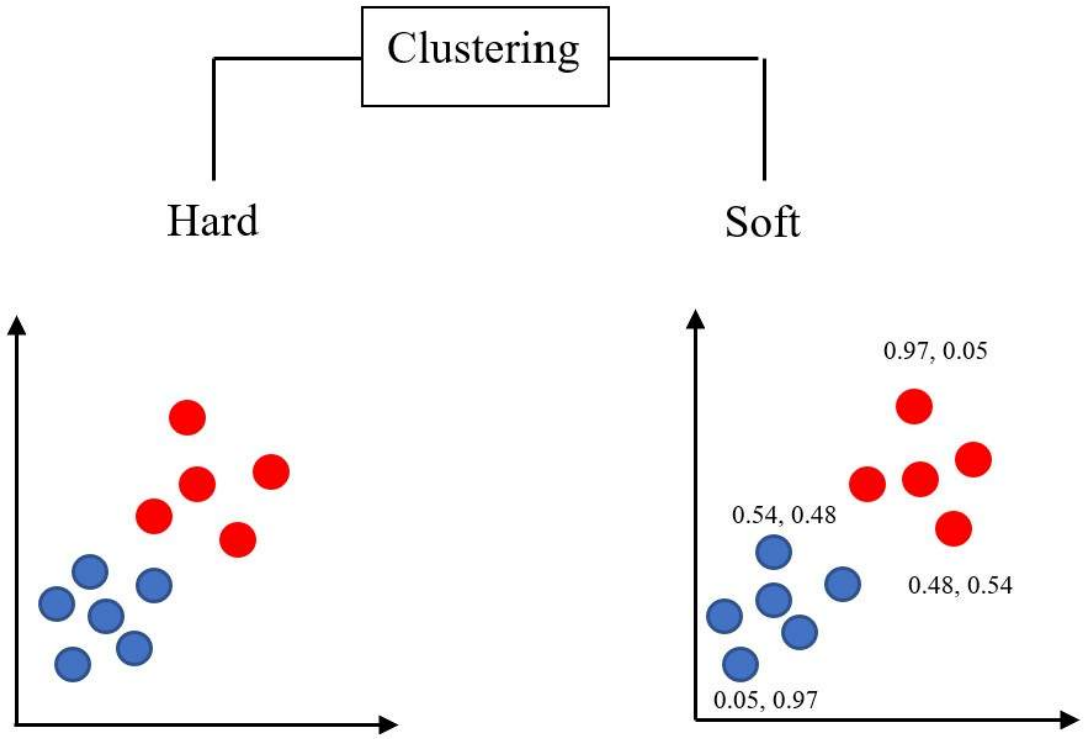


Figure 8.01. Graphical representation of the difference between hard and soft clustering by Arden University.

Continue

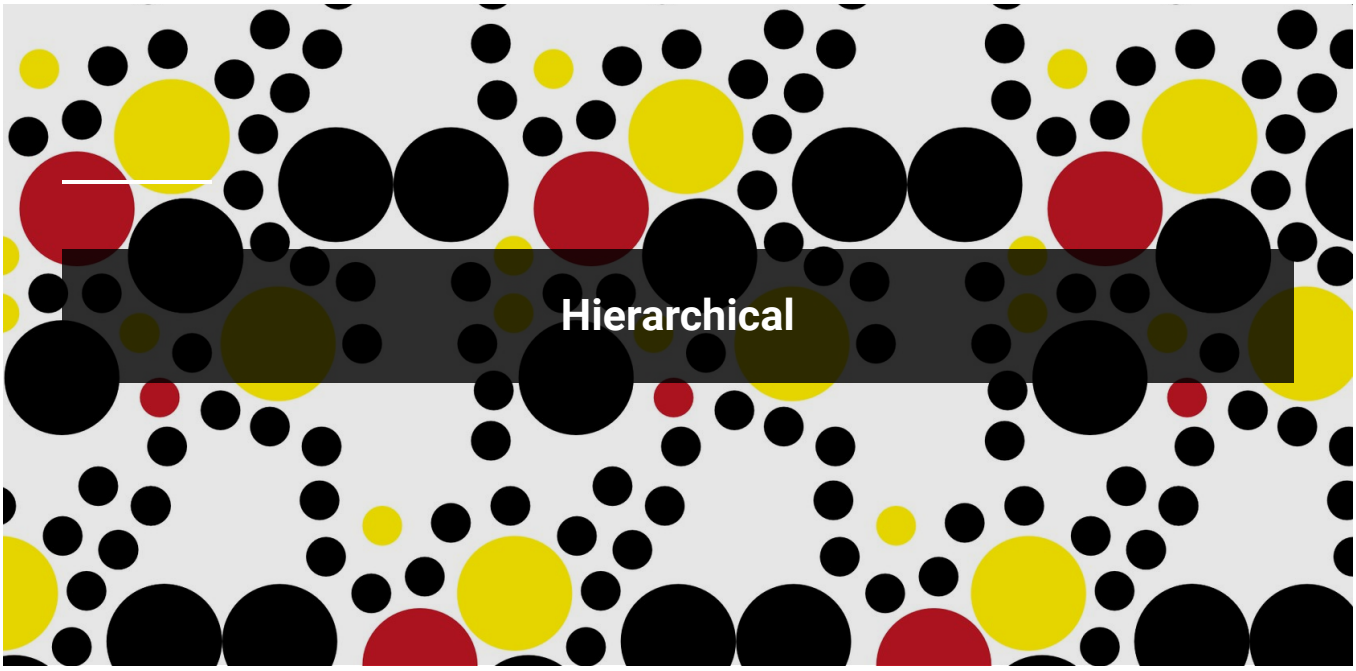


Image from Articulate Rise 360 Content Library.

Using this clustering method produces a hierarchical tree, commonly referred to as a Dendrogram. This particular clustering technique is so widely adopted because there is no requirement to specify the number of clusters from the outset, as the Dendrogram algorithm does this automatically. In turn makes the result from the initial analysis relatively easy to understand.

As can be expected from this clustering method, there are many different real-world applications in which it is applied: business, where customers are separated according to spending habits, or company employees, based upon wages or days taken on sick leave. Hierarchical clustering can also be used for image processing, where handwriting characters are grouped in text recognition. You have probably seen this many times when using an electric note pad when signing for a delivery. In addition to the applications just detailed, there is also information retrieval from previous search results, based upon past queries. A graphical representation of Hierarchical clustering can be seen in Figure 8.02 below.

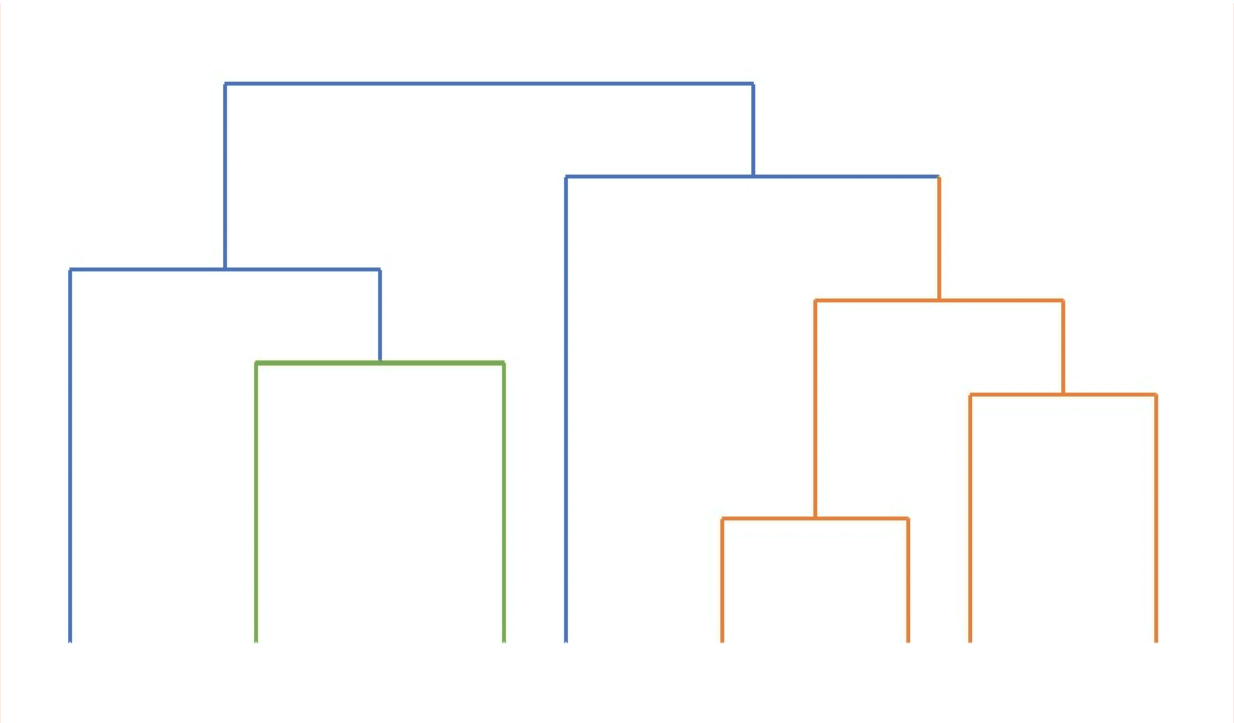


Figure 8.02. An arbitrary hierarchical clustering result by Arden University.

Continue

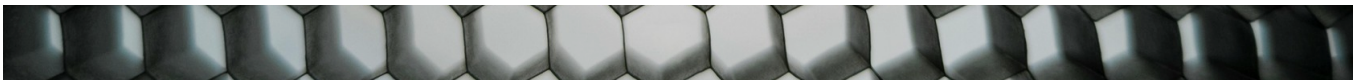


Image from Articulate Rise 360 Content Library.

DBSCAN is an acronym for Density Based Spatial Clustering Application of Noise. This particular clustering algorithm is used when the conventional KM algorithm would not be the best choice to use, when faced with a dataset depicted in Figure 8.03.

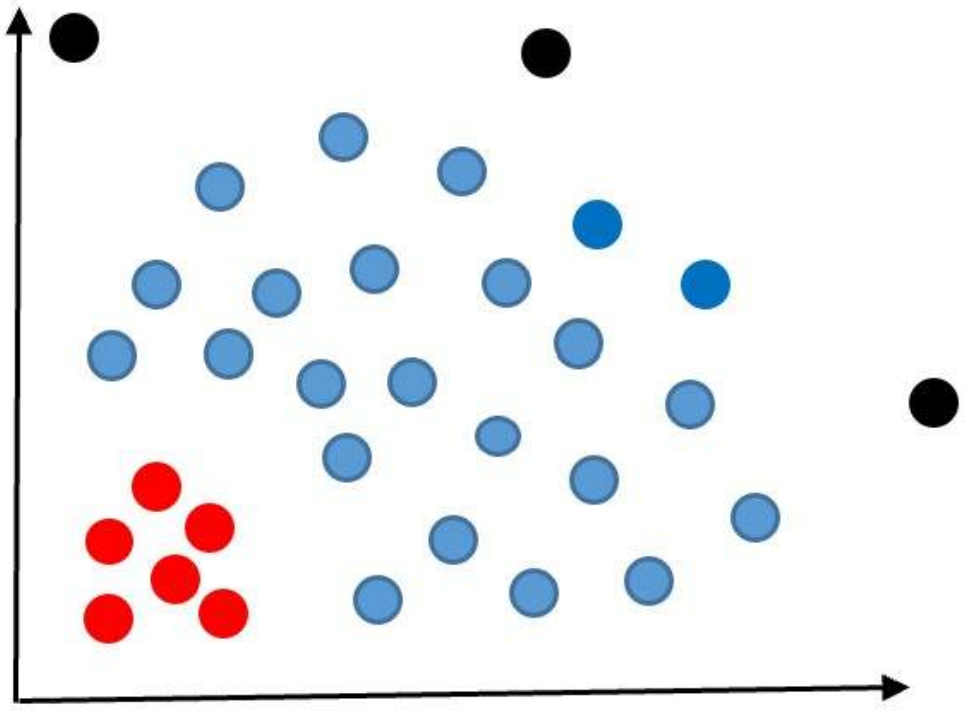


Figure 8.03. Dataset where fixed centroids would not be the ideal technique to use by Arden University.

The dataset which could produce this type of environment is referred to as “nested”. A term used when one obvious cluster wraps around another. Even though the DBSCAN algorithm is used primarily in higher dimensional space, in other words additional features to the data, anything more

that 3D becomes very confusing, so we shall only be studying DBSCAN in 2D space within this topic.

To the naked eye, we can clearly see two separate clusters and some outliers within Figure 8.01. This might be the data collected on peoples' height and weight, for example. Clustering, when using DBSCAN, identifies clusters as high density regions and outliers as low-density regions. What DBSCAN attempts to do is mimic the raw data to help cluster nested data frames. It does this by choosing an arbitrary data point and defining a distance around that point. The other data points which are included within the distance from the starting point are assigned to that point, as Figure 8.04 demonstrates. This process is repeated multiple times until the optimum clustering has been achieved. The main difference between DBSCAN and KM is that the centroids for KM are fixed in place, with DBSCAN they are not. This means that when new data points are introduced to the dataset, the entire DBSCAN algorithm has to be run again, KM does not.

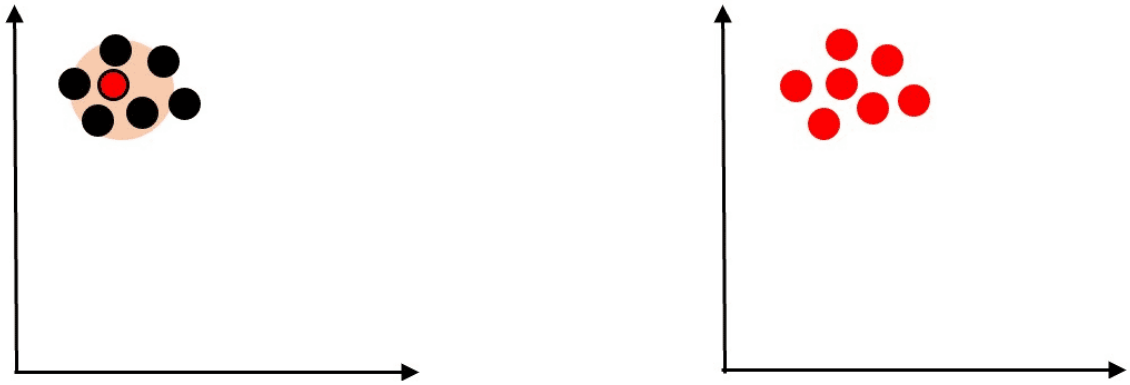


Figure 8.04. Assigning a specified distance from a core point to cluster multiple data points together by Arden University.

Continue

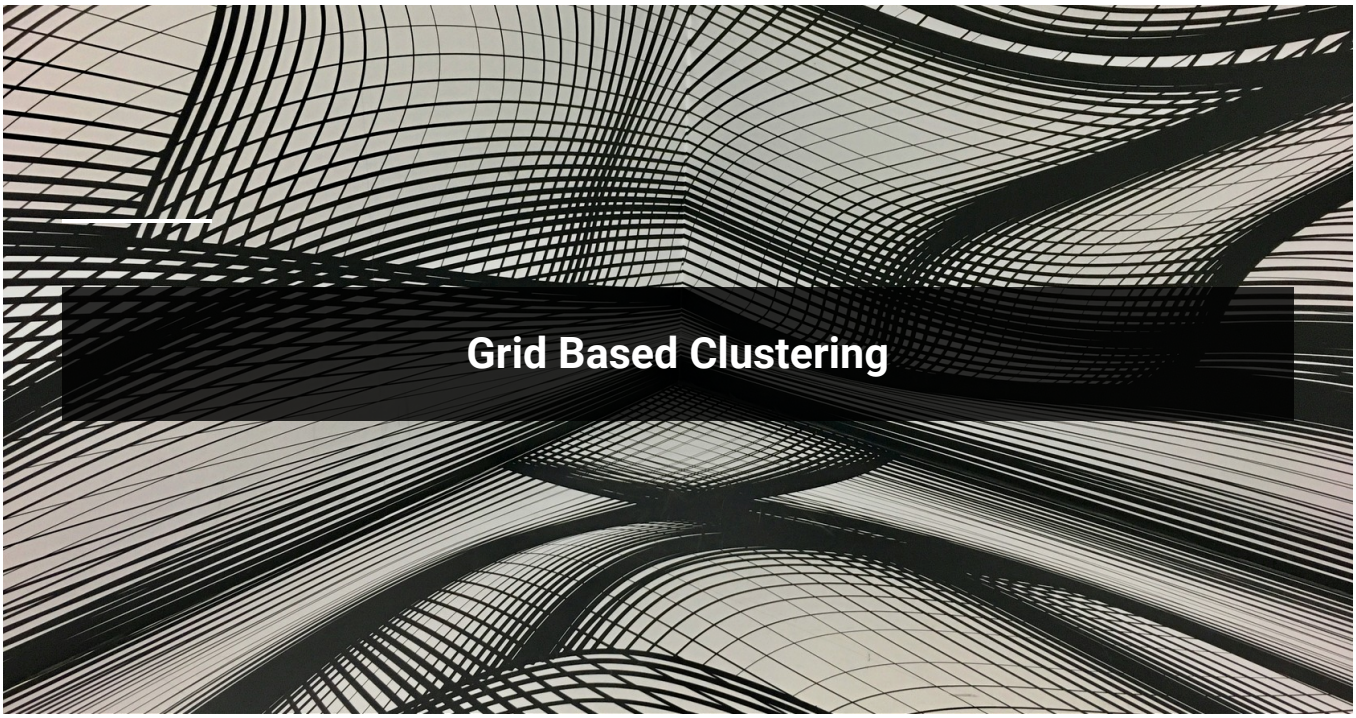


Image from Articulate Rise 360 Content Library.

As the title of this particular method suggests, the algorithm works upon a grid system theoretically imposed upon the data. The benefit of this clustering technique is that the processing time is significantly reduced, in comparison to Hierarchical clustering for example. Grid partitioning uses the Statistical Information Grid approach, or STING for short. STING explores the statistical data identified in each grid cell. As can be determined by such a grid bases approach, there are multiple different levels which correspond to different grid resolutions. Statistical data regarding each cell, such as mean, maximum and minimum values are transformed, pre-determined and stored before the gid is implemented.

A graphical representation of a Grid clustering application can be seen below in Figure 8.05.

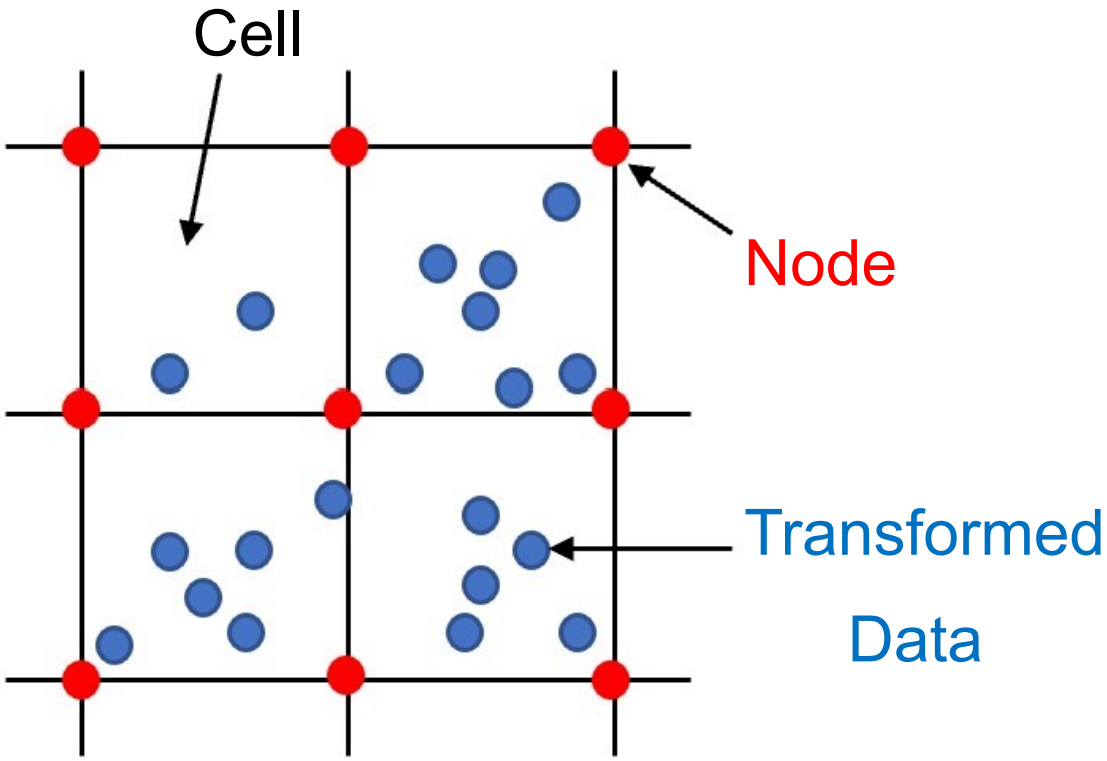
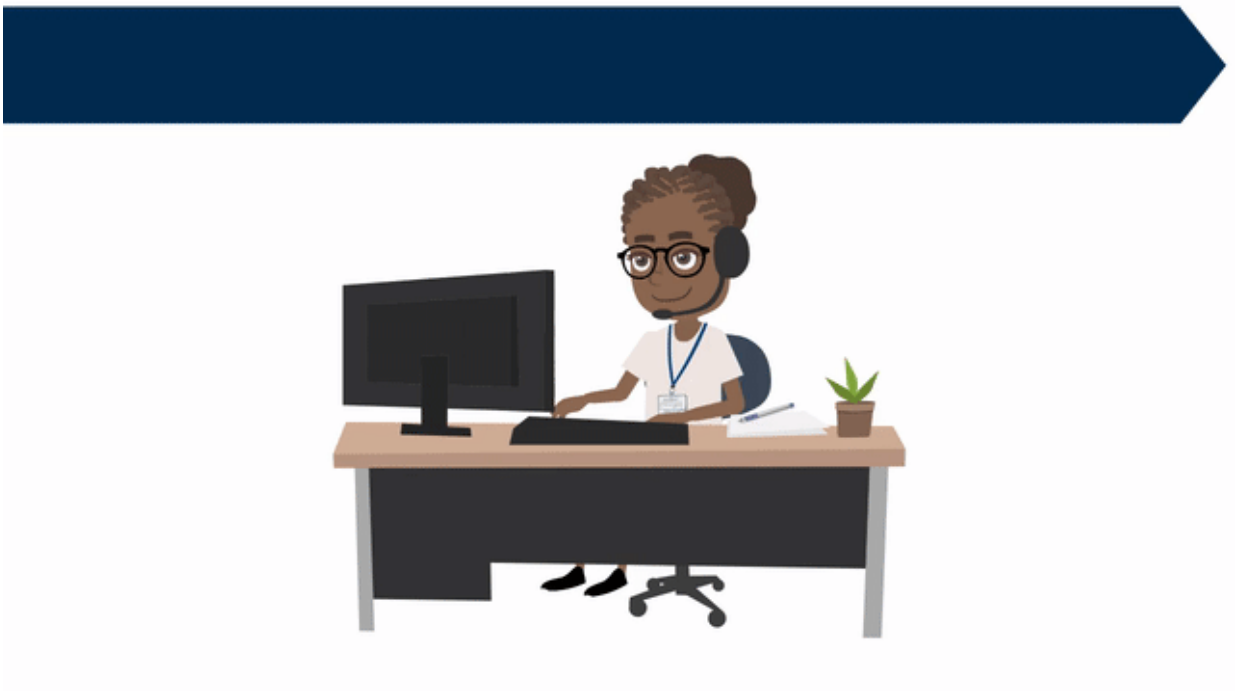


Figure 8.05. An arbitrary representation of a Grid Based cluster process by Arden University. (Accessible text).

Continue

Topic 2 - Activity 1

DBSCAN clustering algorithm



Create a DBSCAN clustering algorithm from 500 randomly generated data points, run the algorithm and print the visualisation. How accurate is the model?

How could you improve the results of the algorithm? What parameters would you change for improvements? Make the changes and run the algorithm again. Have your changes improved the accuracy?

Continue to solution

DOC

Topic 2 activity DBSCAN clustering.docx

354.5 KB

Continue

Topic 2 - Activity 2

Reading

- Chapter 3: Classification: Basic Concepts and Techniques (pp. 133-212) from Introduction to Data Mining, Global Edition by Tan et al.
- Chapter 3: The Data Effect: A Glut at the End of the Rainbow (pp. 126-145) from Predictive Analytics: The Power to Predict Who Will Click, Buy, Lie, or Die, Revised and Updated by Siegel.

Continue

TOPIC 2 SUMMARY

Contained within Topic 2 are the most common clustering techniques and their differing associated approaches. We initial began the topic by determining the clear difference between hard and soft clustering. In doing so that allowed us to progress further into Hierarchical clustering and its disadvantage concerning computational time. DBSCAN was the following on clustering technique available, followed by a brief explanation on a lesser-known clustering technique, namely Grid Clustering.

As you can determine, there are many different techniques available to the machine analyst, dependent upon environment, data and investigation. We shall now take a deep dive into arguably the most common cluster algorithm, K – Means.

Continue to Topic 3

Topic 3: K-Means Clustering

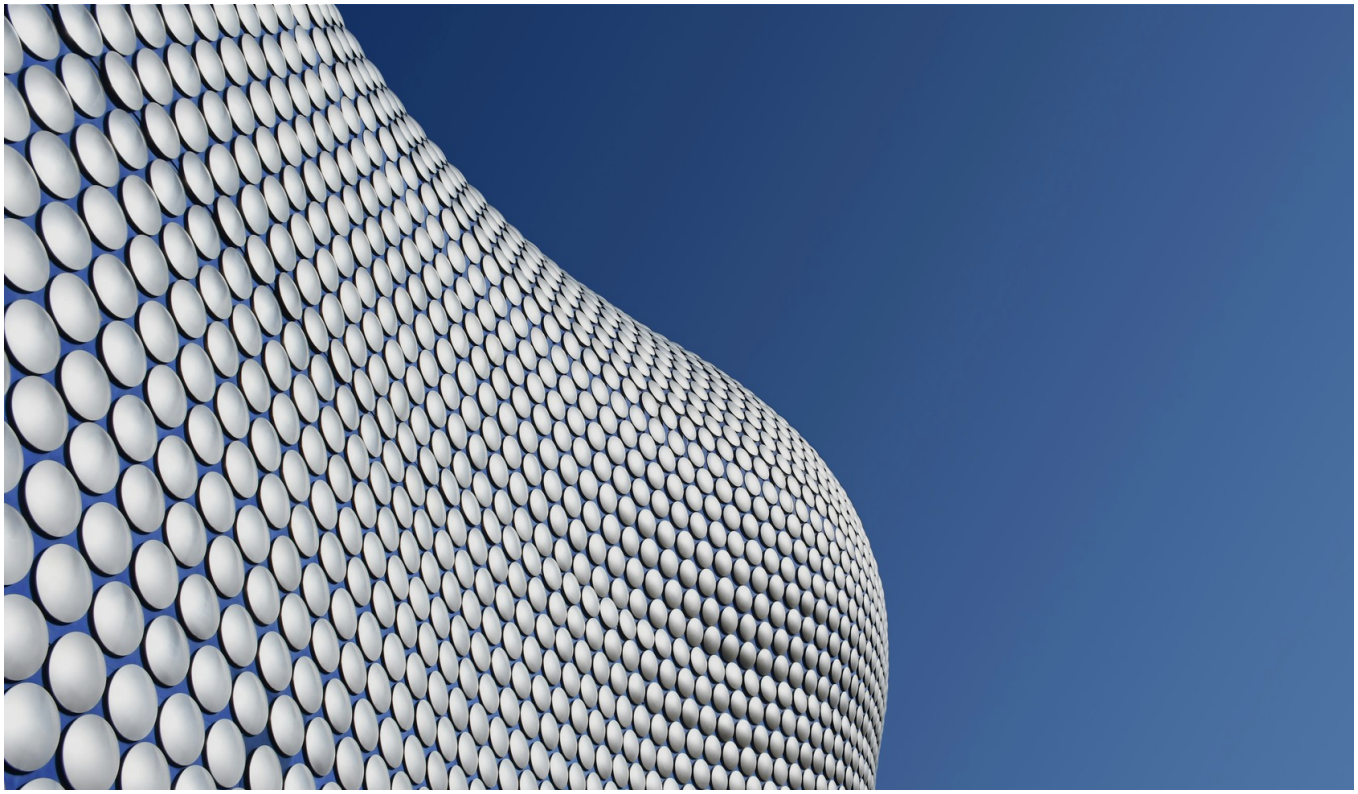


Image from Articulate Rise 360 Content Library.

K – Means (KM) is widely used within industry and many real-world business environments, including:

- **Academic processes**

- Students are categorized, dependent upon their individual grades.

- **Diagnostic systems**
  - The medical industry extensively uses KM to improve digital support systems in the detection of many patients' ailments.

- **Search engines**
  - When a search is undertaken, the search results are grouped together to give associated results. This is exactly what is going on if you perform a new genre search in Netflix, based upon past history.

- **Wireless sensor networks**
  - The cluster algorithm plays a key role in finding cluster heads, which in turn collect the data from the respective clusters.

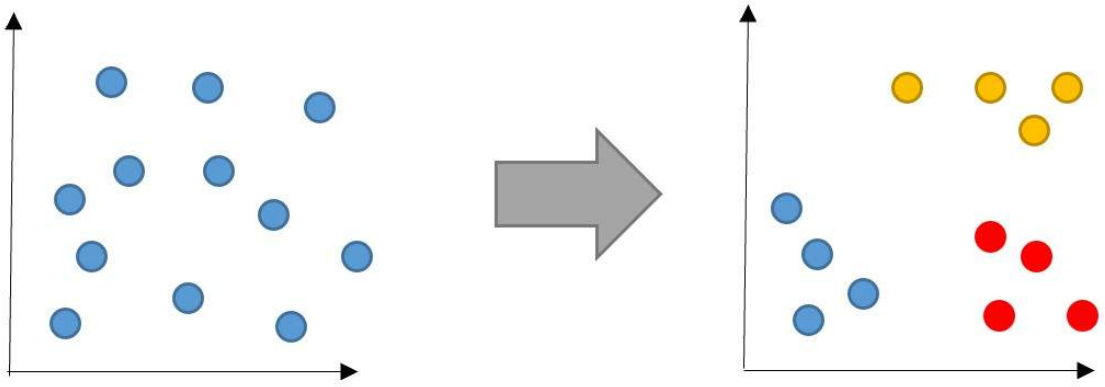


Figure 8.06: Cluster partitioning by Arden University.

As we have hopefully determined from a previous topic, K – Means clustering is different to Hierarchical clustering. Hierarchical informs the analysist what two data points are similar. K –

Means, however, specifically attempts to put the data into the number of clusters the analysist tells it to, in an unsupervised manner.

Say for example you have a large dataset, in which you know nothing about it. However, you want to know if there are any data values that have some type of commonalities. This is where unsupervised partitioning cluster algorithms, i.e., KM comes in quite helpful. As the algorithm will, through multiple iterations, result in separating data values and cluster them together, based upon common characteristics.

As can be conceived, there are numerous metrics that need to be evaluated within a K – Means clustering algorithm. One of the most important metrics we shall highlight here, beginning with Inertia. Basically, the inertia informs us how far apart the data points are within any given cluster. In essence, inertia determines the distance from each data point away from the given centroid. The common method for measuring this is Euclidean distance, but this will only be applicable if the data is numeric. If the data is not numeric, in other words categorical, then the measuring distance employed by K – Means will be Manhattan distance, you are encouraged to research the latter yourself for a well-rounded understanding of cluster measurable distances. As an analyst, you will have to measure each data point distance from its associated centroid. The results are all then summed up to find the final inertia value. Also referred to as the intra-cluster distance. Ideally, as we want the data points to be as similar as possible, the lower the inertia value within each cluster is the good.

When conducting K-Means (KM) clustering, the first step is to initially guess how many centroids (centres) you want to begin with, this is where the K comes in. Say we wanted to begin with three centroids, this would inevitably be K = 3. Note that they are random points and can be initiated anywhere within the dataset. The next step would be to determine the distance you want to set from each centroid to other data points. The shortest distance is selected and then that data point gets assigned to that centroid. Once the algorithm has completed its iterations, the result could be something very similar to Figure 8.06. The variance is then determined and the centroids are again randomly altered in position, the same process is then repeated multiple times, randomly altering centroid points at each iteration. The algorithm then stops.

The optimum centroids for a particular study is determined using the elbow point method. The is where the different variances are plotted, after each centroid change. The optimum point is where the rate in change of the variances begin to gradually decrease as the number of centroid increases. This is shown on Figure 8.07 below.

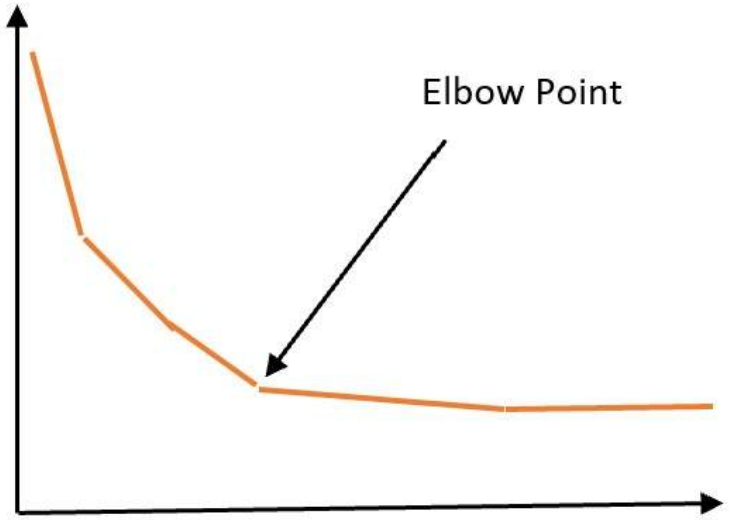


Figure 8.07. Elbow point of variances giving optimal centroid points by Arden University.

K – Means clustering is an ideal clustering application when you have not reach reached a desired outcome, which is usually at the very beginning of studying most large, unlabelled datasets. The big advantage to K – Means clustering is that it is much faster, concerning computational time, then Hierarchical clustering. It is easy to implement and can work on a large number of variables, plus the algorithm can be changed relatively quickly when evaluating the optimum centroid.

Now that you now fully understand what K – Means clustering is and why it is so widely used within a Machine Learning environment, in the next topic we shall be giving step-by-step example of how to implement a K – Means clustering algorithm and the interpretation of such.

Continue

Topic 3 - Activity 1

K – Means

Further research K – Means clustering and state in the group's discussion forum the disadvantages of using the model upon any given dataset. Do the disadvantages discovered also apply to other clustering algorithms?

Continue

Topic 3 - Activity 2

Reading

- Chapter 4: Algorithms: The Basic Methods (pp. 107-151) from Data Mining: Practical Machine Learning Tools and Techniques by Witten et al.
- Chapter 5: Univariate Statistical Analysis (pp. 131-147) from Data Mining and Predictive Analytics by Larose and Larose.
- Chapter 4: The Machine That Learns: A Look Inside Chase's Prediction of Mortgage Risk (pp. 146-183) from Predictive Analytics: The Power to Predict Who Will Click, Buy, Lie, or Die, Revised and Updated by Siegel.
- Chapter 3: Big Data for Business Intelligence (pp. 67-102) from Business Intelligence and Big Data: Drivers of Organizational Success by Olszak.

Continue

TOPIC 3 SUMMARY

Within the third topic, we looked at probably the most common type of clustering algorithms, KM. KM is a portioning clustering technique and as such, pre-define centroids are input into the model, along with the distance from each centroid to the data points. Admittedly, KM is purely guesswork at the beginning of the study, when defining how many centroids to use, but hopefully as the topic progressed you determined that there is a much more statistical test that can be applied to reach optimum clustering, i.e., the elbow method.

In the final topic, we shall be building a K – Means cluster model and evaluating the results.

See you there!

Continue to Topic 4

Topic 4: Building and evaluating a K-Means Clustering model

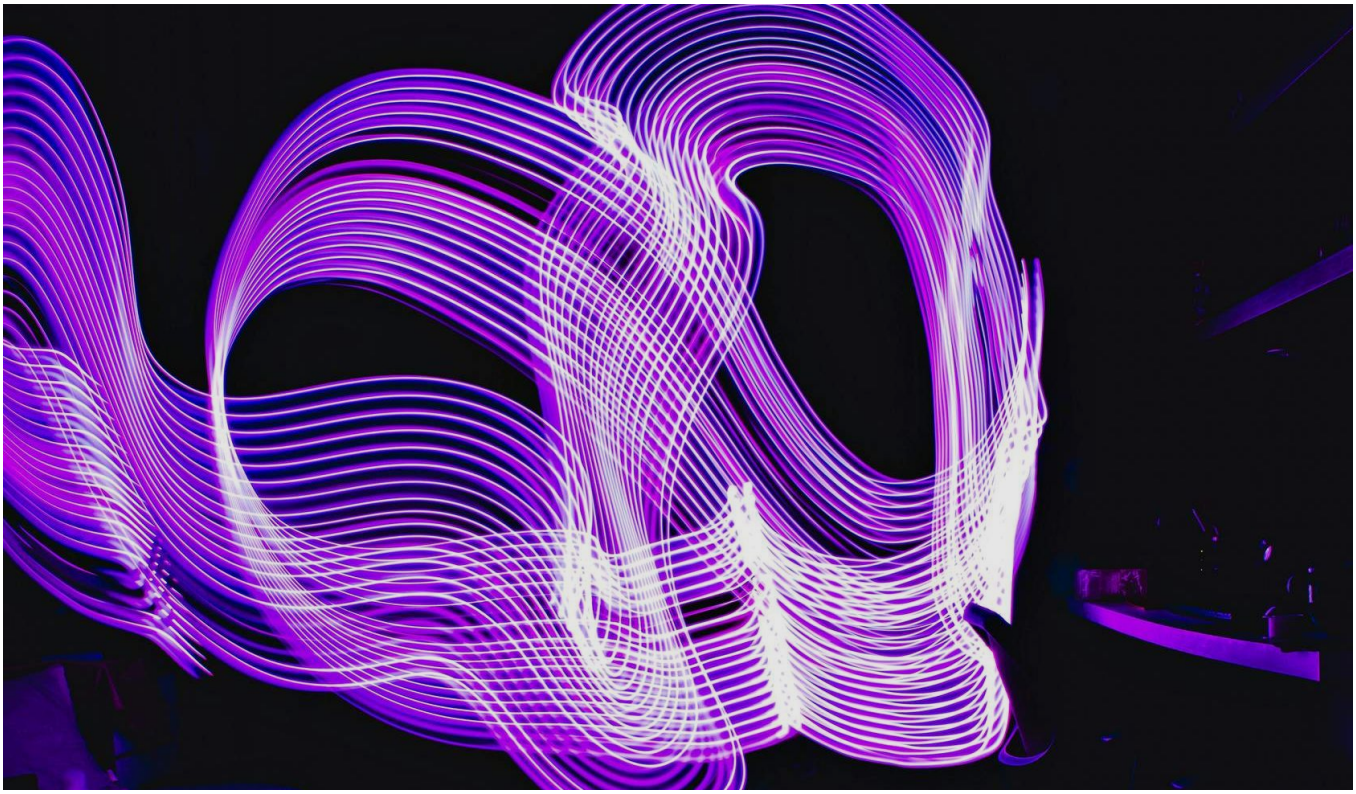
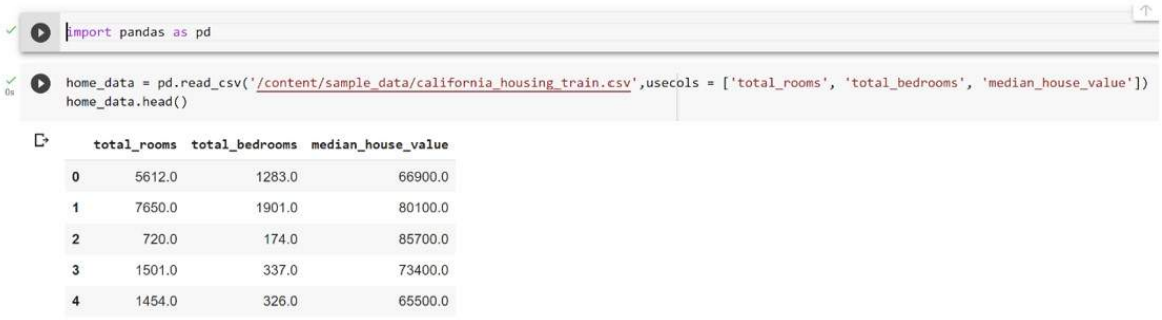


Image from Articulate Rise 360 Content Library.

We shall now run through an example using Google Colab's sample dataset, **california_housing_train**, implementing different centroids, so that we can determine and more importantly compare the differing results and identify an elbow point. We shall be taking the variables, *total rooms*, *total bedrooms* and *median house* value to determine whether there is some type of similarity between them.

First, we need to import the sample data and use only the variables detailed above. Figure 8.08 demonstrates this action being implemented.



```
import pandas as pd
```

```
home_data = pd.read_csv('/content/sample_data/california_housing_train.csv',usecols = ['total_rooms', 'total_bedrooms', 'median_house_value'])
home_data.head()
```

	total_rooms	total_bedrooms	median_house_value
0	5612.0	1283.0	66900.0
1	7650.0	1901.0	80100.0
2	720.0	174.0	85700.0
3	1501.0	337.0	73400.0
4	1454.0	326.0	65500.0

Figure 8.08. Importing the relevant library and sample dataset from Google Colab's sample library in Google Colab by Arden University. (Accessible code).

It is advantageous at this point to plot the data using a scatter plot, with colour gradient, to visually see the variables and how they are represented. This initial scatter plot will be very useful to the analysis as we progress through because we can make a comparison between Figure 8.09 and the resulting, trained cluster algorithm at the end. Figure 8.09 demonstrates the original data plots and construction of the scatter plot.

```
import seaborn as sns

sns.scatterplot(data = home_data, x = 'total_rooms', y = 'total_bedrooms', hue = 'median_house_value')
```

```
<AxesSubplot:xlabel='total_rooms', ylabel='total_bedrooms'>
```

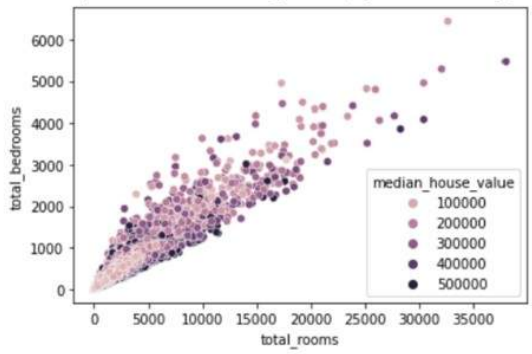


Figure 8.09. Importing the seaborn library and creating the initial scatter plot of targeted variables in Google Colab by Arden University. (Accessible code).

From the initial scatter plot, we can clearly see that on the whole, the less bedroom and total room, the cheaper the house is. However, there are some definitive outliers within the plot, so this makes a compelling argument for us to continue with the study, to try and cluster these variables. However, due to the outliers, a second investigation could be recommended, implementing alternative variables. For this example, however, we shall continue through the process to see what the results give.

The next step is to initialise the test and train split within sklearn using the **train_test_split** function. An important note is that we must always normalise the data when conducting any form of KM clustering investigation. If not, the difference in the variable values will give misleading results and false clusters. Therefore, to normalise the data we need to employ the sklearn library again, using the **preprocessing.normalise** function,as shown in Figure 8.10 below.

```
[6] from sklearn.model_selection import train_test_split

    X_train, X_test, y_train, y_test = train_test_split(home_data[['total_rooms', 'total_bedrooms']], home_data[['median_house_value']], test_size=0.33, random_state=0)

    from sklearn import preprocessing

    X_train_norm = preprocessing.normalize(X_train)
    X_test_norm = preprocessing.normalize(X_test)
```

Figure 8.10. Pre-processing the data through Normalisation in Google Colab by Arden University. (Accessible code).

So far, we have loaded the data and selected only the columns we wish to cluster. We visualised the data using a scatter plot, initialised a test and training split and most importantly, normalised the data. Having now completed the initial setting up we can begin to evaluate the clustering model.

As was mentioned at the beginning of the topic, the number of initial cluster centroids are a complete guess when the first iteration of the algorithm is run, but this is OK because this is where we can evaluate and alter further on, using the elbow method to find the optimum number of centroids. So as our initial guess, we will choose 3, but as I wrote, it could have been 2, or even 5. The initial run is created, then we fit the data to the x and y data labels, as shown below in Figure 8.11.

```
1s  from sklearn.cluster import KMeans

    kmeans = KMeans(n_clusters = 3, random_state = 0, n_init='auto')
    kmeans.fit(X_train_norm)

    KMeans
    KMeans(n_clusters=3, n_init='auto', random_state=0)
```

Figure 8.11. Initialising the clusters and fitting the KM algorithm to x_train in Google Colab by Arden University. (Accessible code).

The initial iteration is now plotted to display our speculative result, shown below in Figure 8.12.

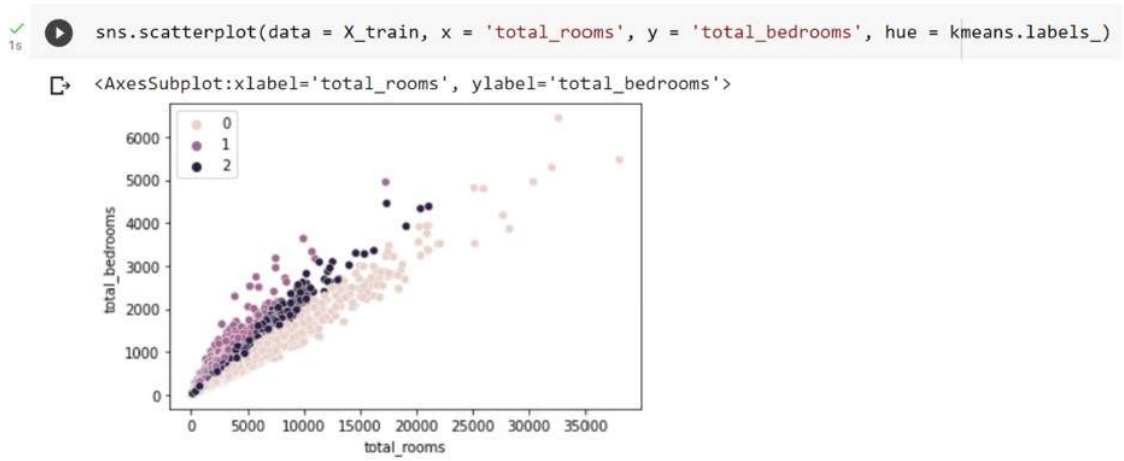


Figure 8.12. Plotting the speculative, initial KM model in Google Colab by Arden University. (Accessible code).

As we can see from Figure 8.12, the data has been definitively clustered into three separate cluster, which is a good result because now we know that the clustering algorithm is working as it should and also there is some form of similarities between the variables. Plus, by running an evaluation metric on the clustering algorithm, we can further determine whether the model is a good fit. With this evaluation metric, the lower the score, the better the fit.

Now here comes the fun part: we do not know whether the three cluster we initially guessed is a good fit, so further iterations of the model need to be run, altering the cluster numbers each time to eventually find the optimum number of clusters. This can of course be written into the code to be done automatically, however here we will do it manually, so that it is easier to see the result of increasing the number of clusters. So, we shall begin with two clusters and plot the results. The code and first result are shown in Figure 8.13.

```
K = range(2, 8)
fits = []
score = []


for k in K:
    model = KMeans(n_clusters = k, random_state = 0, n_init='auto').fit(X_train_norm)

    fits.append(model)

    score.append(silhouette_score(X_train_norm, model.labels_, metric='euclidean'))
```

Figure 8.13. The results of running the second iteration with K = 2 in Google Colab by Arden University. (Accessible code).

```
sns.scatterplot(data = X_train, x = 'total_rooms', y = 'total_bedrooms', hue = fits[0].labels_)
```

```
<AxesSubplot:xlabel='total_rooms', ylabel='total_bedrooms'>
```

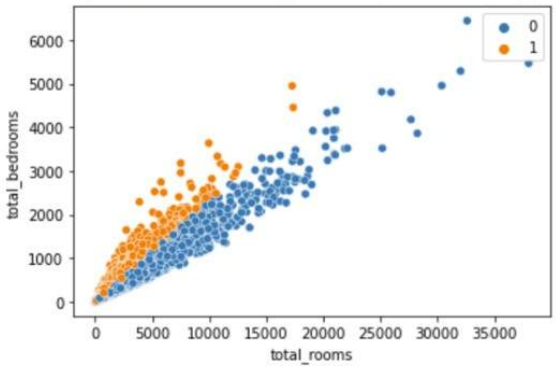


Figure 8.13b. The results of running the second iteration with K = 2 in Google Colab by Arden University. (Accessible code).

Now let's run the model again, altering the number of clusters each time to include 3, 4, 5 and 7 clusters. The comparative results are shown in Figure 8.14.

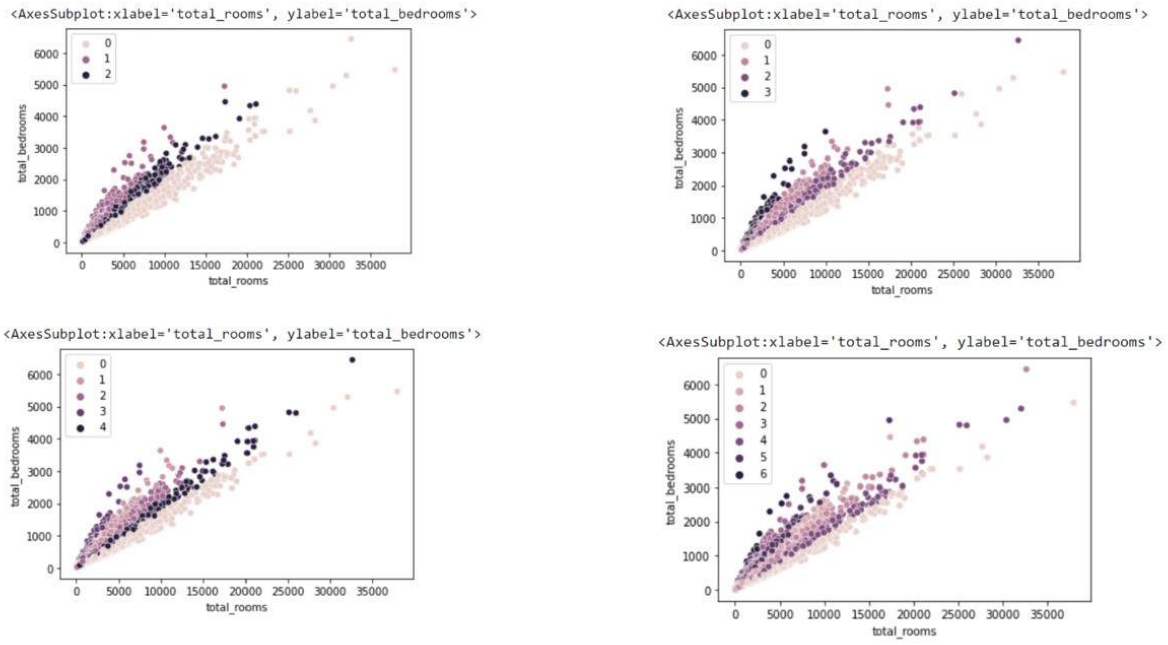


Figure 8.14. Comparative results of altering K to find the optimal clustering in Google Colab by Arden University. (Accessible code).

We can clearly see from Figure 8.14 that the more clusters we add, the more the model appears to degenerate. However, this is just a snap observation and we need to approach the optimisation in a more mathematical manner, through the elbow method, as demonstrated in Topic 3. Generally, as we increase the number of clusters the model does improve, until a point is reached where increasing the number of clusters does not improve the model at all. This optimum point can be plotted on a simple 2D plane, using the aforementioned elbow method. The derivation and result of all tested cluster results detailed in Figure 8.14 can be shown below in Figure 8.15.

```
sns.lineplot(x = K, y = score)
```

```
<AxesSubplot:>
```

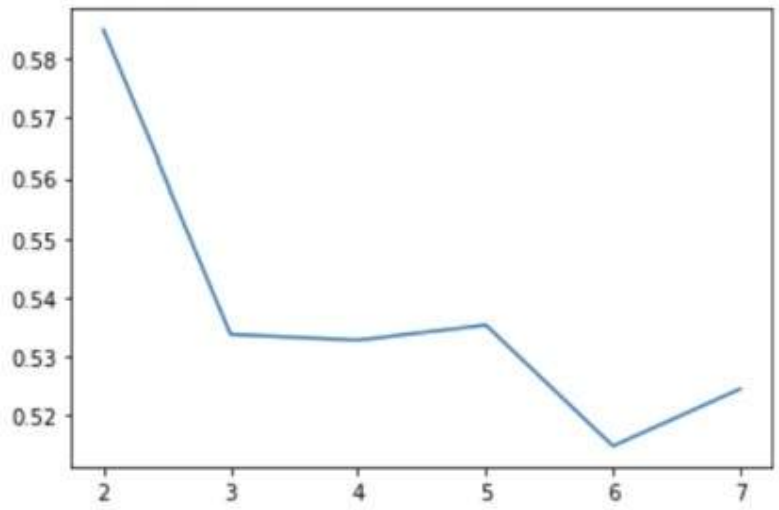


Figure 8.15. Plotting the variances to evaluate the optimum K in Google Colab by Arden University. (Accessible code).

If we look carefully at Figure 8.15, we can see that there is a sharp decline from 1 – 2 clusters, then the line plot evens itself out between clusters 3 – 4 and then becomes very erratic between clusters 4 – 7. This clearly informs us that the optimum number of clusters for this particular investigation is 3.

To summarise, we loaded the data and selected only the variables we wished to cluster. We then visualised the data using a scatter plot, initialised a test and training split and most importantly, normalised the data. From that point, we began by running the KM clustering algorithm and altering the number of clusters. Finally employed the elbow method to give the optimum number of clusters for this study. As a final comparative to fully realise the benefits of running the KM algorithm we shall show Figures 8.09 and 8.12 collectively as Figure 8.16.

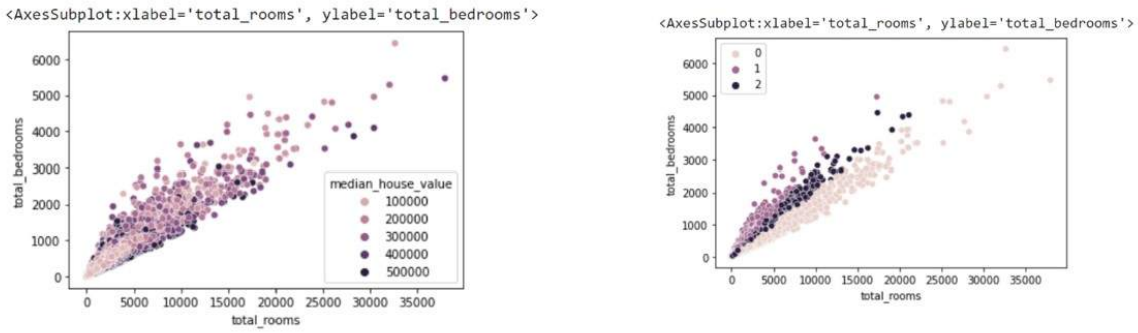
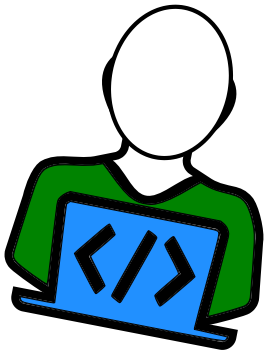


Figure 8.16. Comparative clustering model in Google Colab by Arden University.

Continue

Topic 4 - Activity 1

Create a KM clustering algorithm from 100 randomly generated data points, run the algorithm and print the visualisation. How accurate is the model?



How could you improve the results of the algorithm? What parameters would you change for improvements? Make the changes and run the algorithm again. Have your changes improved the accuracy? Post your solutions in the discussion forum for further analysis by the tutor and your peers.

Continue to solution

DOC **Topic 4 activity KM clustering.docx**
776.5 KB

Continue

Topic 4 - Activity 2

Reading

- Chapter 4: Association Analysis: Basic Concepts and Algorithms (pp. 213-306) from Introduction to Data Mining, Global Edition by Tan *et al.*
- Chapter 5: Univariate Statistical Analysis (pp. 131-147) from Data Mining and Predictive Analytics by Larose and Larose.
- Chapter 5: Model Evaluation and Improvement (pp. 184-226) from Introduction to Machine Learning with Python: A Guide for Data Scientists by Müller and Guido.
- Chapter 3: The Data Effect: A Glut at the End of the Rainbow (pp. 126-145) from Predictive Analytics: The Power to Predict Who Will Click, Buy, Lie, or Die, Revised and Updated by Siegel.

Continue

TOPIC 4 SUMMARY

Well done, you have completed Topic 4.

In the final topic we have demonstrated a K – Means cluster investigation on a targeted dataset. As you can hopefully determine, the study is involved and care must be taken to ensure that baseline results are compared against processed results and you are encouraged to follow this approach when conducting your investigation. In doing so, you will have demonstrated the improvements made to the model in terms of accuracy.

All topics included within this module have now been concluded.

Continue to Knowledge check

Knowledge check

Select the correct answer to the following questions.

1 of 3: In DBSCAN, what will increasing the distance around a data point do?

- It will usually reduce the resulting number of clusters.
- It will usually increase the resulting number of clusters.
- Nothing at all.

SUBMIT

2 of 3: Does the Ward linkage do?

- The Ward approach utilises the minimum distances of the clusters.
- The Ward approach utilises the average distances of the clusters.
- The Ward approach utilises the variances of the clusters, rather than measuring the distances directly.

SUBMIT

3 of 3: Is the following statement **true** or **false**?

Hierarchical clustering uses pre-defined centroids.

- True
- False

SUBMIT

Continue to Lesson summary

Lesson summary

LESSON SUMMARY

Well done, you have completed Lesson 8 on K – Means Cluster analysis.

Contained within Topic 1 is a detailed theoretical explanation of what clustering is and more importantly why it is used so extensively within a machine learning environment. Although cluster analysis would not be used as a standalone technique, it is nevertheless a very good starting point on which to impose upon an unlabelled dataset. Using a cluster analysis as an initial starting point, it gives analysts some guidance on the next steps to take for a more focused study.

Topic 2 detailed the most common clustering techniques and their differing associated approaches. We initial began by determining the clear difference between hard and soft clustering and in doing so, allowed us to progress further into Hierarchical clustering, density-based clustering and the overlooked clustering technique, Grid clustering.

Within Topic 3, we focused in on arguably the most common type of clustering algorithms, K – Means. As we are now aware, K – Means is a portioning clustering technique and as such, pre-define centroids are input into the model, along with the distance from each centroid to the data points. Hopefully, as the topic progressed you determined that there is a much more statistical test that can be applied to reach optimum clustering, i.e., the elbow method.

In the final topic we demonstrated a K – Means cluster investigation on a targeted dataset. The study is involved and care must be taken to ensure that baseline results are compared against processed results in terms of accuracy.

Lesson 8 on Cluster analysis is now completed. In the following lesson we shall be taking yet another deep dive into a popular, alternative form of cluster analysis, namely Hierarchical clustering.

See you there.

Continue to 'Reading list'

Reading List

All reading and resources in this module are provided by the Arden University Library service. The full reading list for this module can be accessed at the link below. More books, journals and databases can be found on the Library Portal. If you need these resources in a different format, our accessibility tools can help.

Full reading list

You can find all the key, essential and wider reading and resources for this module on the dedicated reading list page on the Learning Success Hub.

VIEW READING LIST